

4×1 HDMI Switcher Design for Multimedia Applications

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Abstract—This project presents the design of a 4×1 HDMI switcher using the TMDS461 HDMI switch IC. The system enables switching between four HDMI input sources and routing the selected signal to a single HDMI output display. The design includes TMDS differential signal routing, DDC communication handling using SDA and SCL lines, and hot-plug detection support. A push-button controlled selection mechanism is implemented using counter-based control logic. The complete schematic and PCB layout were designed in KiCad while maintaining high-speed signal integrity considerations such as differential routing and controlled impedance. The proposed design provides an efficient and low-cost HDMI switching solution suitable for multimedia applications.

GitHub Repository: HDMI-Switcher-Design

Index Terms—HDMI Switcher, TMDS Signaling, HDMI Routing, Differential Signaling, Multimedia Switching, KiCad PCB Design, DDC Communication

I. INTRODUCTION

HDMI (High-Definition Multimedia Interface) has become one of the most common standards for transferring digital audio and video between electronic devices such as televisions, computers, gaming consoles, and display monitors. It is capable of carrying high-quality multimedia data through a single cable while maintaining reliable communication at very high speeds. HDMI technology uses Transition Minimized Differential Signaling (TMDS) to transmit data efficiently with lower noise and reduced electromagnetic interference, making it suitable for high-resolution video applications.

In many practical situations, users often need to connect several HDMI source devices to a single display. Continuously unplugging and reconnecting cables is not only inconvenient but can also damage connectors over time. To solve this problem, HDMI switchers are used to select one source from multiple HDMI inputs and direct it to a single output display without requiring manual cable replacement.

Designing an HDMI switcher is not straightforward because HDMI signals operate at very high frequencies. Maintaining proper signal integrity is important to ensure stable video transmission. Factors such as differential pair routing, impedance control, electrostatic discharge protection, hot-plug detection, and DDC communication through SDA and SCL lines must be carefully considered during the design process.

Improper PCB routing or signal handling can lead to data loss, display instability, or communication failure.

The work presented in this project focuses on the design of a 4×1 HDMI switcher using the TMDS461 HDMI switch IC. The system is capable of selecting one HDMI input from four available sources and transmitting it to a single HDMI output. The design also incorporates TMDS differential signal routing, DDC line management, and push-button controlled input selection. Both the schematic design and PCB layout were developed in KiCad while following recommended high-speed PCB design practices for HDMI-based systems.

II. DESIGN COMPONENTS

A. HDMI Connector and Signal Description

The HDMI connector serves as the interface between the multimedia source devices and the HDMI switcher system. In this project, HDMI Type-A connectors are used for four input ports and one output port. HDMI supports high-speed digital transmission of audio, video, and control information through multiple dedicated signal lines. The HDMI interface mainly consists of TMDS data channels, TMDS clock channel, DDC communication lines, Hot Plug Detect (HPD) signal, and power lines. The signals D0, D1, and D2 represent the TMDS data channels used for high-speed transmission of digital video information. Each channel consists of a differential pair. These differential pairs carry encoded video data using Transition Minimized Differential Signaling. Differential signaling improves noise immunity and reduces electromagnetic interference, enabling reliable high-speed transmission over HDMI cables. The TMDS channels primarily carry RGB pixel information and control data. Since the TMDS signals operate at multi-Gbps speed, the PCB traces must be routed as controlled impedance differential pairs with proper spacing and length matching. The CK+ and CK- signals form the TMDS clock differential pair. The clock channel provides synchronization information for the TMDS data channels. The receiver uses the clock signal to correctly sample the incoming video data transmitted over D0, D1, and D2 channels. Similar to TMDS data lines, the clock signals are routed as high-speed differential pairs to maintain signal integrity and minimize noise.

The SDA (Serial Data) and SCL (Serial Clock) lines form the Display Data Channel communication interface based on

the I²C protocol. These lines are used for communication between the HDMI source device and the display. Through the DDC channel, the source device reads EDID (Extended Display Identification Data) information from the display. EDID contains supported resolutions, refresh rates, and display capabilities. Proper DDC communication is necessary for successful HDMI operation. The SDA and SCL lines are open-drain signals and require pull-up resistors. In this design, 4.7 kohms pull-up resistors connected to 3.3V are used for reliable I²C communication.

The HPD, Hot Plug Detect signal is used by the display device to indicate its presence to the HDMI source. When a display is connected and ready, the HPD signal becomes active, informing the source device that HDMI communication can begin. After detecting HPD, the source initiates DDC communication and starts TMDS video transmission. In the HDMI switcher system, the HPD signal is managed through the TMDS461 switch IC to ensure that only the selected HDMI input communicates with the output display.

The HDMI connector provides a +5V supply from the HDMI source device. This voltage is mainly used for HDMI detection and control circuitry. In the proposed design, Schottky diodes are used to combine the +5V lines from multiple HDMI inputs while preventing reverse current flow between HDMI ports. The HDMI connector includes multiple ground pins and shield connections to provide signal return paths and reduce electromagnetic interference.

B. TMDS461 HDMI Switch IC and Signal Description

The TMDS461PZTR is a high-speed 4x1 HDMI switch IC used in this project to route HDMI signals from four input ports to a single HDMI output port. The IC supports Transition Minimized Differential Signaling, which is the signaling technique used in HDMI communication for reliable high-speed transmission of digital video and audio data. The TMDS461 acts as the central switching element of the HDMI switcher system and allows selection of one HDMI input source at a time. In addition to switching high-speed TMDS video signals, the IC also manages Display Data Channel communication, Hot Plug Detect signals, and HDMI detection functionality. The TMDS461 contains four HDMI input channels and one HDMI output channel. Each HDMI input consists of four differential signal pairs: D0+, D0-, D1+, D1-, D2+, D2-, CLK+, and CLK-. The D0, D1, and D2 channels are TMDS data channels used for transmitting encoded video information. These differential pairs primarily carry RGB pixel data and synchronization information between the HDMI source and display device. The CLK differential pair carries the TMDS clock signal used for synchronization of the incoming TMDS data streams. Since these signals operate at multi-Gbps speed, they must be routed as controlled impedance differential pairs on the PCB. Proper spacing, trace width control, and length matching are required to maintain signal integrity and reduce electromagnetic interference.

The TMDS461 internally routes one selected HDMI input channel to the output sink channel. The output side

of the IC contains TMDS sink signals such as D0_SINK+, D0_SINK-, D1_SINK+, D1_SINK-, D2_SINK+, D2_SINK-, CLK_SINK+, and CLK_SINK-. These signals are connected to the HDMI output connector and transmit the selected HDMI signal to the display device. The switching operation is controlled using the S0 and S1 selection pins of the TMDS461. Depending on the logic levels applied to these pins, one of the four HDMI input channels is selected. In this project, the select signals are generated using a push-button controlled counter circuit. The logic combinations 00, 01, 10, and 11 correspond to HDMI Input 1, HDMI Input 2, HDMI Input 3, and HDMI Input 4 respectively.

The TMDS461 also manages DDC communication through the SDA and SCL lines. SDA represents the Serial Data line, while SCL represents the Serial Clock line of the I²C-based Display Data Channel (DDC). These lines are used for communication between the HDMI source and display device. Through DDC communication, the source reads EDID (Extended Display Identification Data) information from the display, which contains supported resolutions, refresh rates, and display capabilities. Since SDA and SCL are open-drain signals, 4.7 kohms pull-up resistors connected to 3.3V are used to establish proper logic HIGH levels and ensure reliable communication.

Another important signal managed by the TMDS461 is the Hot Plug Detect (HPD) signal. HPD is used by the display device to indicate that a valid HDMI display connection is present. When HPD becomes active, the HDMI source begins DDC communication and subsequently initiates TMDS video transmission. The TMDS461 routes the HPD signal only to the selected HDMI source, ensuring proper communication between the active source and the output display.

The IC also contains 5V_PWR and 5V_SINK pins associated with HDMI detection and control circuitry. The 5V_PWR pins are connected to the +5V signals received from the HDMI input connectors, while the 5V_SINK pin is associated with the HDMI output side. In this design, Schottky diodes are used to combine multiple HDMI +5V sources while preventing reverse current flow between HDMI ports. This arrangement protects the HDMI sources and allows safe power sharing.

The TMDS461 operates using a 3.3V supply and contains multiple VCC and GND pins distributed around the IC package to support stable operation during high-speed switching. To improve power integrity and reduce switching noise, 0.1μF decoupling capacitors are connected close to each VCC pin. Additional bulk capacitors are placed near the power entry region of the IC to stabilize the supply voltage. The VSADJ pin of the TMDS461 is used to control the TMDS output voltage swing, and a resistor connected between VSADJ and ground sets the required output amplitude for HDMI compliance.

C. Power Supply Circuit Using AMS1117

The power supply section of the proposed HDMI switcher is designed to provide a stable 3.3V supply required for the operation of the TMDS461 HDMI switch IC and associated

control circuitry. The circuit consists of four Schottky diodes, an AMS1117-3.3 voltage regulator, and input/output filtering capacitors. The HDMI standard provides a +5V signal from each HDMI source device through pin 18 of the HDMI connector. In this design, the +5V signals from the four HDMI input ports are labeled as VI_1, VI_2, VI_3, and VI_4. The four Schottky diodes are connected to these HDMI +5V input lines to implement a power OR-ing configuration. The primary purpose of the Schottky diodes is to allow any active HDMI source to supply power to the circuit while preventing reverse current flow between HDMI ports. Without these diodes, connecting multiple HDMI sources directly together could cause backfeeding, resulting in source conflicts or possible damage to connected devices. Schottky diodes are specifically selected because they have a low forward voltage drop compared to conventional silicon diodes. Typically, Schottky diodes exhibit a forward voltage drop of approximately 0.2V to 0.3V, which helps maintain sufficient voltage at the regulator input. This is important because the AMS1117 voltage regulator requires an adequate input voltage to produce a stable regulated 3.3V output.

The combined output of the Schottky diode network is connected to the input of the AMS1117-3.3 low-dropout voltage regulator. The AMS1117 converts the incoming +5V supply into a regulated 3.3V output used to power the TMDS461 IC and other low-voltage circuitry. The regulator provides stable voltage operation and helps reduce the effect of supply fluctuations from the HDMI source devices. Since the TMDS461 operates using a 3.3V supply, proper voltage regulation is essential for reliable HDMI signal switching and communication. Capacitors C3 and C4 are used for voltage stabilization and noise filtering around the AMS1117 regulator. Capacitor C3, connected at the input side of the regulator, acts as an input filter capacitor and helps reduce voltage ripple and transient noise present on the incoming supply line. Capacitor C4 is connected at the output side of the regulator and improves output voltage stability while reducing noise on the regulated 3.3V rail. These capacitors also help improve transient response during rapid current changes in the HDMI circuitry.

The output of the AMS1117 regulator is labeled as 3p3V and is distributed to the TMDS461 HDMI switch IC, DDC pull-up resistors, and other supporting circuitry. Proper grounding is maintained throughout the power supply section to reduce noise coupling and improve overall power integrity. Since HDMI systems involve high-speed TMDS differential signaling, stable and low-noise power delivery is critical to maintain reliable video transmission and prevent signal degradation.

D. Input Selection Circuit Using 74HC192 Counter

The HDMI input selection mechanism in the proposed HDMI switcher is implemented using a push-button controlled counter circuit based on the 74HC192 synchronous up/down counter IC. The purpose of this circuit is to generate digital selection signals for the TMDS461 HDMI switch IC, allowing

TABLE I
COMPONENTS USED IN THE HDMI SWITCHER DESIGN

S. No.	Components Used	MPN (DigiKey)
1	Switch IC	TMDS461PZTR
2	3.3V Voltage Regulator	AMS1117-3.3
3	4 Bit Counter	74HC192
3	HDMI Type A Connector	SS-53000-001
5	Push Button Switch	1301.9314.24

the user to switch between four different HDMI input sources using a single push button. The counter circuit provides a simple hardware-based input selection method without requiring a microcontroller.

The 74HC192 is a 4-bit synchronous up/down counter capable of generating binary count values at its output pins. Although the IC supports four-bit counting, only the two least significant output bits, QA and QB, are used in this project because the TMDS461 requires only two selection bits to select one of four HDMI inputs. The output signals QA and QB are connected to the S0 and S1 selection pins of the TMDS461 HDMI switch IC. Depending on the binary combination generated by QA and QB, one HDMI input channel is selected. The logic combinations 00, 01, 10, and 11 correspond to HDMI Input 1, HDMI Input 2, HDMI Input 3, and HDMI Input 4 respectively.

The push-button switch SW1 is used to generate manual clock pulses for the counter circuit. Whenever the push button is pressed, a clock pulse is applied to the UP clock input of the 74HC192 counter, causing the counter to increment its output state. As the count value changes, the QA and QB outputs also change, resulting in the selection of the next HDMI input channel. This arrangement allows sequential switching between the four HDMI inputs through repeated button presses.

There is a pull-up resistor connected to the push-button input line. The pull-up resistor ensures that the input signal remains at a stable logic HIGH level when the push button is not pressed. Without this resistor, the input line could float and produce unstable or unpredictable logic levels. When the push button is pressed, the input line is connected to ground, generating a logic LOW pulse required for counter operation. The RC debounce network filters the rapid fluctuations and produces a cleaner clock pulse for reliable counting operation. Capacitor temporarily stores charge and smooths the signal transitions, while resistor limits current flow and controls the charging and discharging time constant of the debounce network. The CLEAR pin of the 74HC192 is connected to a reset network consisting of capacitor and resistor. This RC network provides power-on reset functionality for the counter circuit. When power is initially applied, capacitor creates a temporary reset condition that initializes the counter outputs to a known state. This ensures that the HDMI switcher starts with a predictable input selection condition during system startup. Resistor provides a discharge path for the capacitor and helps establish stable reset timing behavior.

The LOAD, DOWN, and unused counter output pins are

connected to fixed logic levels to ensure proper counter operation and prevent floating inputs. Since the design only requires upward counting for sequential HDMI input selection, the UP counting mode of the 74HC192 is used while the DOWN counting functionality remains disabled. The counter circuit operates using the regulated 3.3V power supply generated by the AMS1117 voltage regulator. Proper grounding and stable supply voltage are necessary to ensure reliable logic transitions and accurate HDMI input selection.

III. POWER PATH

The power path of the system begins at the HDMI input connectors. According to the HDMI standard, each HDMI source device provides a +5V supply through pin 18 of the HDMI connector. In this design, the +5V signals from the four HDMI input ports are labeled as VI_1, VI_2, VI_3, and VI_4. These four power inputs are connected to a Schottky diode network. The Schottky diodes perform power OR-ing functionality, allowing any active HDMI source to provide power to the system while preventing reverse current flow between different HDMI ports. This arrangement protects connected devices from backfeeding and ensures safe power distribution.

The combined output of the Schottky diode network is connected to the AMS1117-3.3 voltage regulator. The AMS1117 converts the incoming +5V supply into a stable regulated 3.3V output required by the TMDS461 HDMI switch IC and the 74HC192 counter circuit. Capacitors connected at the input and output of the regulator provide voltage filtering, noise reduction, and transient stabilization. The regulated 3.3V rail is distributed throughout the circuit to power the HDMI switching IC, DDC pull-up resistors, counter logic, and associated control circuitry. Additional 0.1 μ F decoupling capacitors are placed near the VCC pins of the TMDS461 to suppress high-frequency switching noise and improve power integrity during HDMI operation.

IV. SIGNAL PATH

The signal path of the design consists of high-speed TMDS differential signals, DDC communication lines, and HDMI control signals. Each HDMI input connector provides three TMDS data differential pairs (D0 $\pm$, D1 $\pm$, D2 $\pm$) and one TMDS clock differential pair (CLK $\pm$). The TMDS signals enter the TMDS461 HDMI switch IC. The internal ESD protection circuitry of TMDS protects the high-speed HDMI lines from electrostatic discharge and voltage spikes that may occur during cable insertion or handling. Inside the TMDS461, one HDMI input channel is selected based on the control signals applied to the S0 and S1 select pins. The selected TMDS differential pairs are internally routed to the TMDS sink outputs connected to the HDMI output connector. This allows only one HDMI source to communicate with the output display at a time. Since TMDS signals operate at multi-Gbps speeds, the PCB traces carrying these signals are routed as controlled impedance differential pairs with careful length matching and minimal vias to preserve signal integrity.

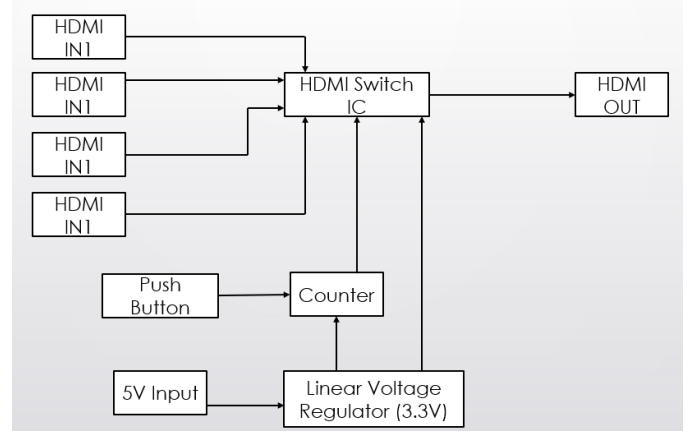


Fig. 1. Block Diagram

In addition to the TMDS video signals, the signal path also includes DDC communication lines and HDMI control signals. The SDA and SCL lines implement the Display Data Channel communication interface based on the I²C protocol. These lines are switched through the TMDS461 so that the selected HDMI source can communicate with the output display and read EDID information such as supported resolutions and refresh rates. Pull-up resistors connected to the 3.3V supply establish stable logic HIGH levels on the SDA and SCL lines. The Hot Plug Detect signal forms another important part of the control signal path. The display device asserts the HPD signal to indicate that a valid HDMI connection is present. The TMDS461 routes the HPD signal only to the selected HDMI source, allowing proper synchronization between the active source and the display. The 5V detection lines associated with the HDMI inputs are also connected to the TMDS461 to support HDMI source detection and switching functionality.

The HDMI input selection process is controlled by the push-button and 74HC192 counter circuit. Whenever the push button is pressed, the counter increments its output state and generates new logic levels on the QA and QB outputs. These outputs are connected to the S0 and S1 selection pins of the TMDS461, thereby changing the active HDMI input channel. In this manner, the control path interacts with the signal path to perform HDMI source switching operation.

V. RESULTS

The PCB layout of the HDMI switcher was designed using KiCad. The TMDS461 HDMI switch IC was placed centrally on the PCB to minimize differential signal trace lengths between HDMI input connectors and the HDMI output connector. Decoupling capacitors were placed close to the VCC pins of the TMDS461 to reduce supply noise and improve power integrity. Ground plane implementation and short trace routing techniques were used throughout the PCB design. The dimensions of layout are 85.81 mm X 62.62 mm.

The 3D model of the HDMI switcher PCB was generated using the KiCad 3D visualization environment to verify component placement, connector alignment, and overall mechani-

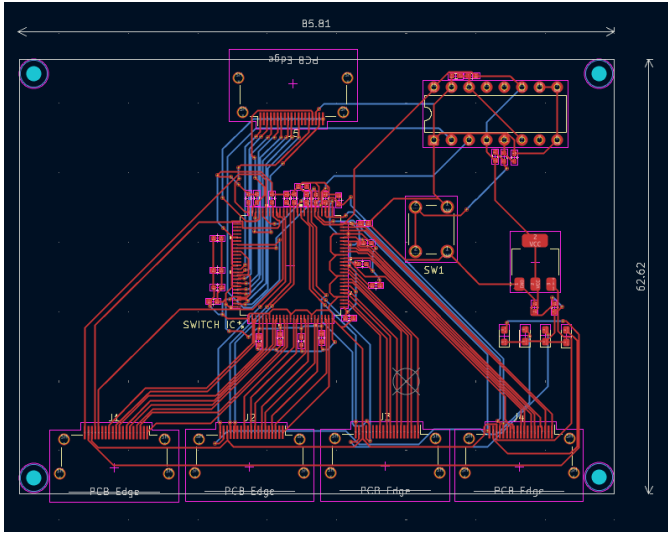


Fig. 2. Layout (without ground pouring)

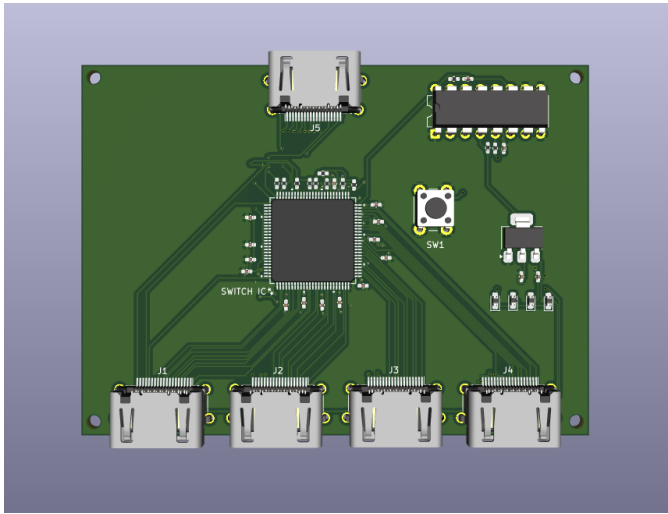


Fig. 3. Proposed 3D Model

cal structure of the design. The 3D model provides a realistic representation of the fabricated PCB and helps identify possible placement conflicts before manufacturing. The generated model also assists in evaluating board dimensions, connector accessibility, and overall packaging feasibility of the HDMI switcher system.

The estimated hardware cost of the proposed 4×1 HDMI switcher was calculated based on component prices obtained from DigiKey. The major components contributing to the overall system cost include the TMDS461 HDMI switch IC, HDMI Type-A connectors, 74HC192 counter IC, push-button switch, and AMS1117-3.3 voltage regulator. Since the design uses four HDMI input ports and one HDMI output port, a total of five HDMI connectors were required. The TMDS461 HDMI switch IC represents the major portion of the overall system cost due to its high-speed HDMI switching functionality. The total estimated component cost of the proposed HDMI

switcher design was approximately INR 1165.57, excluding PCB fabrication and assembly costs. The analysis demonstrates that the proposed design provides a relatively low-cost solution for HDMI source switching applications while maintaining high-speed HDMI signal routing capability.

TABLE II
COST OF COMPONENTS SELECTED FROM DIGIKEY

Component	Price (INR)
HDMI Type-A Connector	73.62
TMDS461PZTR HDMI Switch IC	597.32
74HC192 Counter IC	90.63
Push Button Switch	81.20
AMS1117-3.3 Voltage Regulator	28.32

VI. CONCLUSION

The key contributions of the proposed work is the implementation of a compact and low-cost HDMI switching architecture using dedicated hardware logic without requiring multiple microcontroller-based control system. The use of a push-button controlled counter circuit simplifies the input selection mechanism while reducing system complexity and cost. In addition, the integration of safe power OR-ing circuitry using Schottky diodes improves power reliability and protects HDMI source devices from reverse current flow.

The proposed design demonstrates a practical and scalable solution for multimedia switching applications requiring multiple HDMI inputs and a single display output. Future improvements may include support for higher HDMI standards, automatic source detection, infrared remote control functionality, microcontroller-based intelligent switching, and advanced signal integrity optimization for higher bandwidth HDMI applications.

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